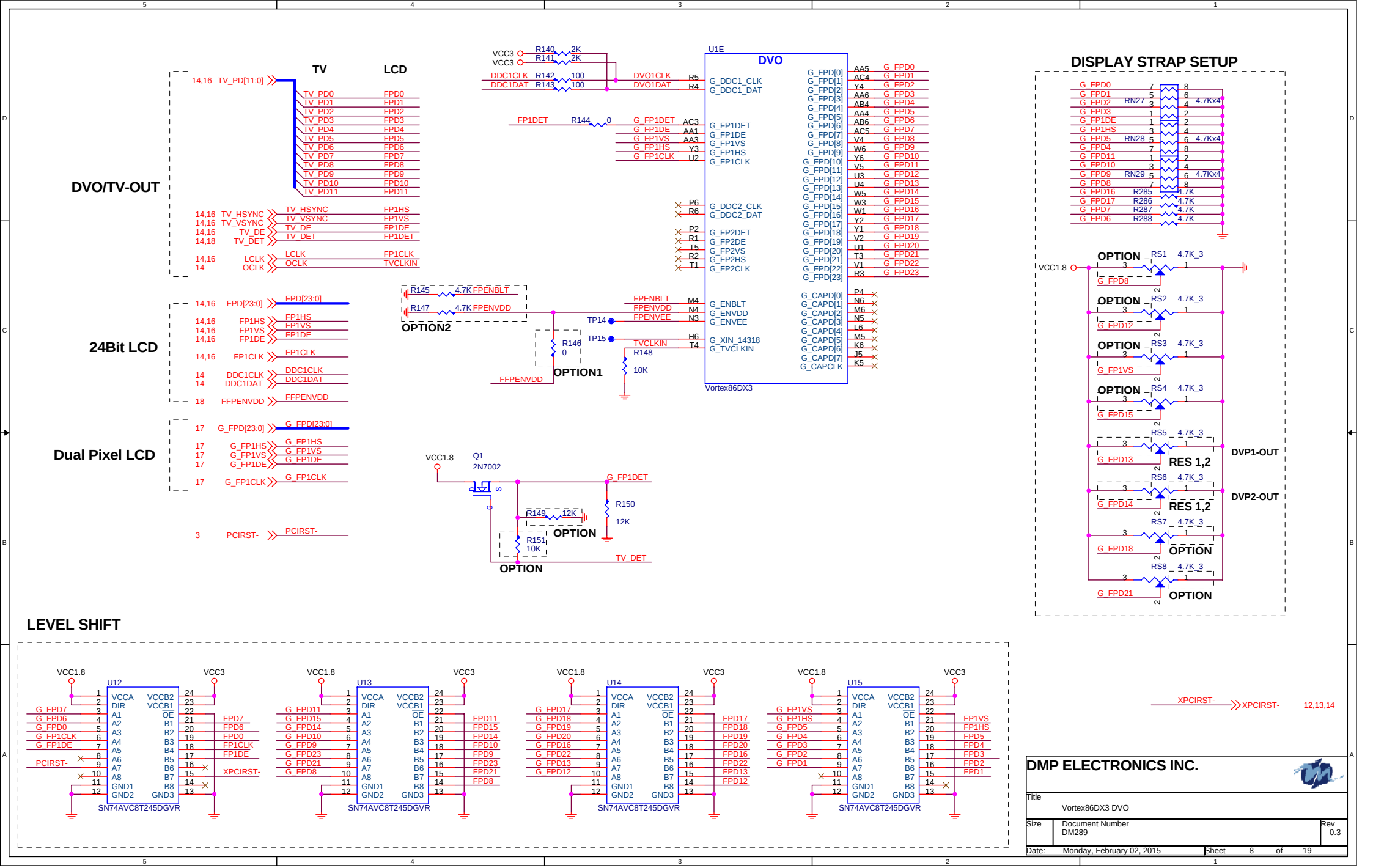
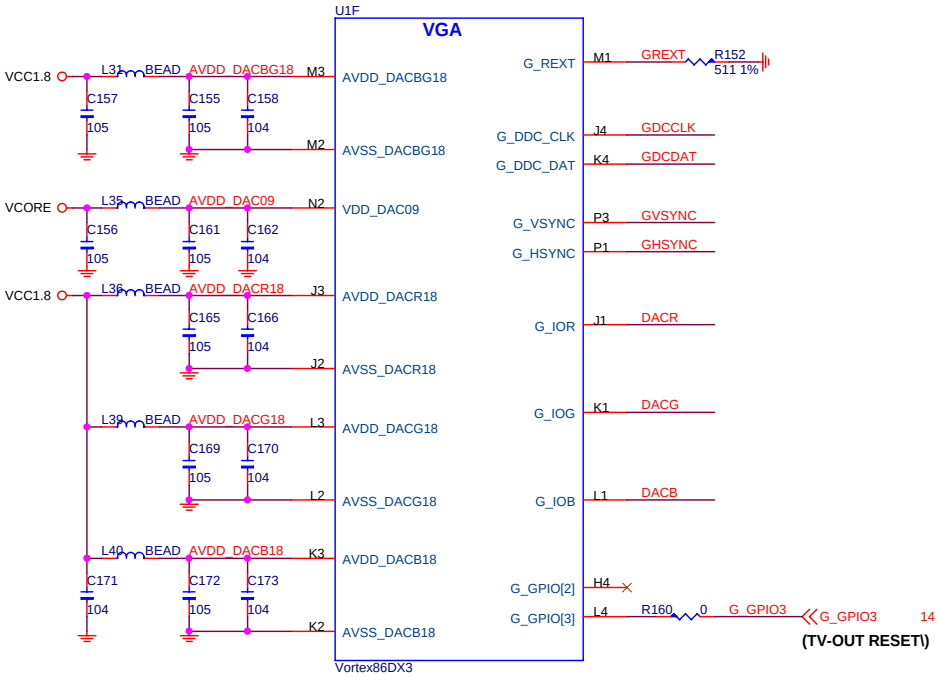
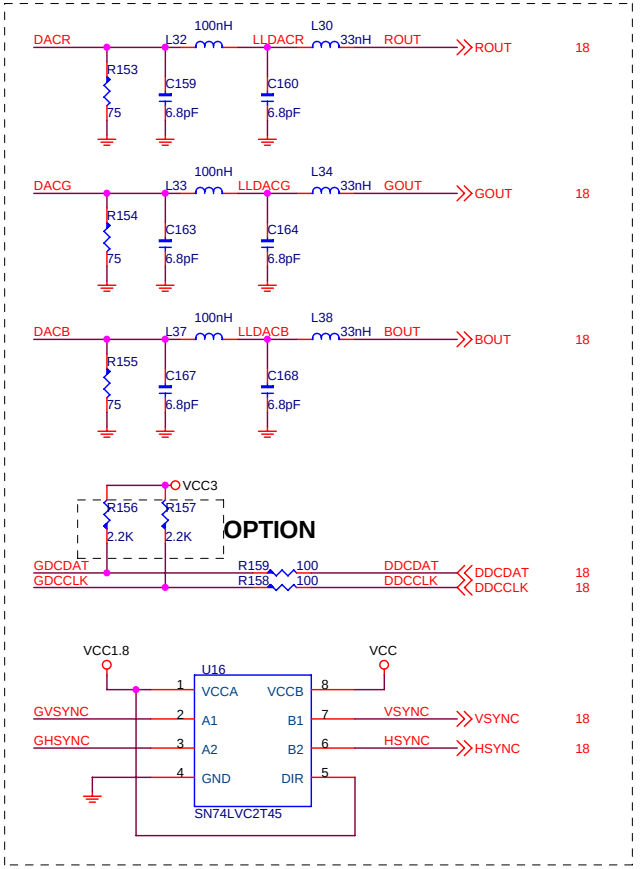

DVO/TV-OUT
TV
LCD
14,16 TV_PD[11:0]
TV_PD0 FPD0
TV_PD1 FPD1
TV_PD2 FPD2
TV_PD3 FPD3
TV_PD4 FPD4
TV_PD5 FPD5
TV_PD6 FPD6
TV_PD7 FPD7
TV_PD8 FPD8
TV_PD9 FPD9
TV_PD10 FPD10
TV_PD11 FPD11
14,16 TV_HSYNC TV_HSYNC FP1HS
14,16 TV_VSYNC TV_VSYNC FP1VS
14,16 TV_DE TV_DE FP1DE
14,18 TV_DET TV_DET FP1DET
14,16 LCLK LCLK FP1CLK
14 OCLK OCLK TVCLKIN
24Bit LCD
14,16 FPD[23:0] FPD[23:0]
14,16 FP1HS FP1HS
14,16 FP1VS FP1VS
14,16 FP1DE FP1DE
14,16 FP1CLK FP1CLK
14 DDC1CLK DDC1CLK
14 DDC1DAT DDC1DAT
18 FFPENVDD FFPENVDD
Dual Pixel LCD
17 G_FPD[23:0] G_FPD[23:0]
17 G_FP1HS G_FP1HS
17 G_FP1VS G_FP1VS
17 G_FP1DE G_FP1DE
17 G_FP1CLK G_FP1CLK
3 PCIRST- PCIRST-
VCC3
R140 2K
R141 2K
DDC1CLK R142 100 DVO1CLK
DDC1DAT R143 100 DVO1DAT
FP1DET R144 0 G_FP1DET
U1E
DVO
G_DDC1_CLK
G_DDC1_DAT
G_FP1DET
G_FP1DE
G_FP1VS
G_FP1HS
G_FP1CLK
G_DDC2_CLK
G_DDC2_DAT
G_FP2DET
G_FP2DE
G_FP2VS
G_FP2HS
G_FP2CLK
G_ENBLT
G_ENVDD
G_ENVEE
G_XIN_14318
G_TVCLKIN
G_FPD[0]
G_FPD[1]
G_FPD[2]
G_FPD[3]
G_FPD[4]
G_FPD[5]
G_FPD[6]
G_FPD[7]
G_FPD[8]
G_FPD[9]
G_FPD[10]
G_FPD[11]
G_FPD[12]
G_FPD[13]
G_FPD[14]
G_FPD[15]
G_FPD[16]
G_FPD[17]
G_FPD[18]
G_FPD[19]
G_FPD[20]
G_FPD[21]
G_FPD[22]
G_FPD[23]
G_CAPD[0]
G_CAPD[1]
G_CAPD[2]
G_CAPD[3]
G_CAPD[4]
G_CAPD[5]
G_CAPD[6]
G_CAPD[7]
G_CAPCLK
Vortex86DX3
R145 4.7K FPENBLT
R147 4.7K FPENVDD
OPTION2
FPENBLT
FPENVDD
FPENVEE
TP14
TP15
TVCLKIN
R148 10K
R146 0
OPTION1
FFPENVDD
VCC1.8
Q1
2N7002
G_FP1DET
R149 12K
OPTION
R150 12K
R151 10K
TV_DET
OPTION
DISPLAY STRAP SETUP
G_FPD0
G_FPD1
G_FPD2
G_FPD3
G_FP1DE
G_FP1HS
G_FPD5
G_FPD4
G_FPD11
G_FPD10
G_FPD9
G_FPD8
G_FPD16
G_FPD17
G_FPD7
G_FPD6
RN27 4.7Kx4
RN28 4.7Kx4
RN29 4.7Kx4
R285 4.7K
R286 4.7K
R287 4.7K
R288 4.7K
VCC1.8
OPTION RS1 4.7K_3 G_FPD8
OPTION RS2 4.7K_3 G_FPD12
OPTION RS3 4.7K_3 G_FP1VS
OPTION RS4 4.7K_3 G_FPD15
RS5 4.7K_3 G_FPD13 RES 1,2 DVP1-OUT
RS6 4.7K_3 G_FPD14 RES 1,2 DVP2-OUT
RS7 4.7K_3 G_FPD18 OPTION
RS8 4.7K_3 G_FPD21 OPTION
LEVEL SHIFT
U12
U13
U14
U15
VCC1.8
VCC3
VCCA
VCCB2
DIR
VCCB1
OE
GND1
GND2
GND3
SN74AVC8T245DGVR
G_FPD7 G_FPD6 G_FPD0 G_FP1CLK G_FP1DE PCIRST-
FPD7 FPD6 FPD0 FP1CLK FP1DE XPCIRST-
G_FPD11 G_FPD15 G_FPD14 G_FPD10 G_FPD9 G_FPD23 G_FPD21 G_FPD8
FPD11 FPD15 FPD14 FPD10 FPD9 FPD23 FPD21 FPD8
G_FPD17 G_FPD18 G_FPD19 G_FPD20 G_FPD16 G_FPD22 G_FPD13 G_FPD12
FPD17 FPD18 FPD19 FPD20 FPD16 FPD22 FPD13 FPD12
G_FP1VS G_FP1HS G_FPD5 G_FPD4 G_FPD3 G_FPD2 G_FPD1
FP1VS FP1HS FPD5 FPD4 FPD3 FPD2 FPD1
XPCIRST- XPCIRST- 12,13,14
DMP ELECTRONICS INC.
Title
Vortex86DX3 DVO
Size
Document Number
DM289
Rev
0.3
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VGA





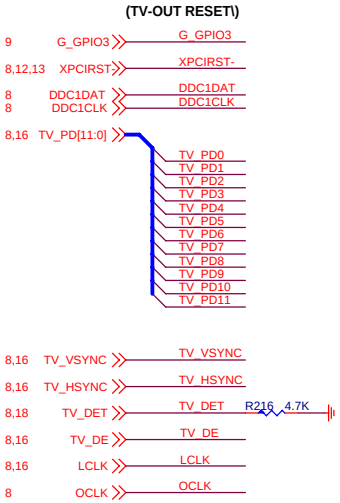
DMP ELECTRONICS INC.		
Title: Vortex86DX3 VGA/ PLL		
Size	Document Number DM289	Rev 0.3
Date: Monday, February 02, 2015	Sheet 9 of 19	

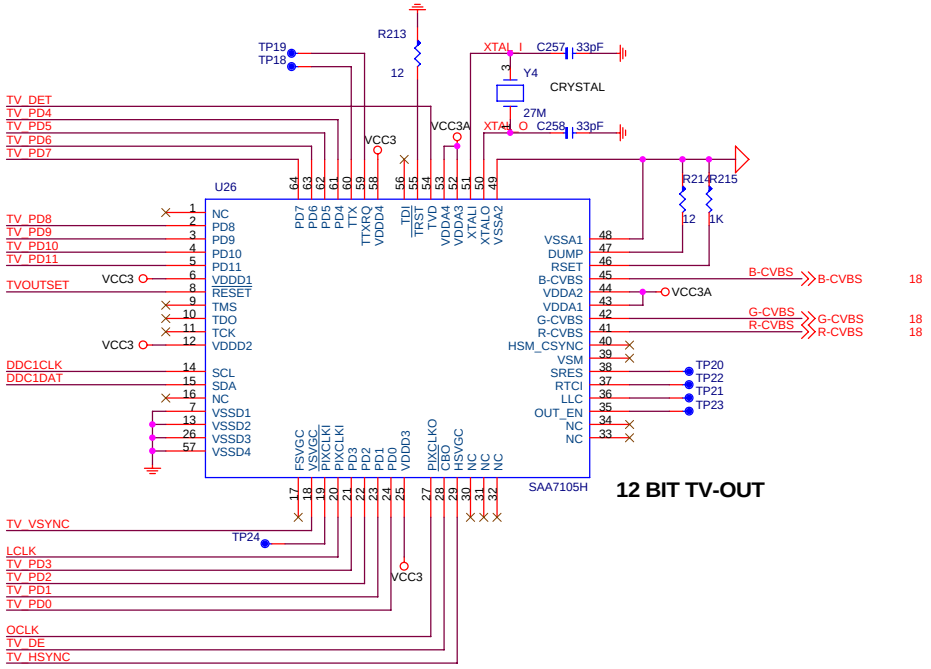
TV-OUT OPTION

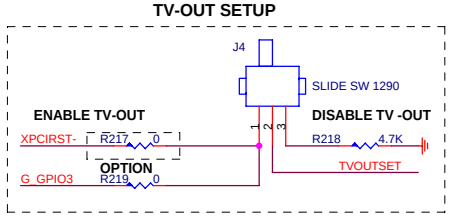
If not on pieces of U28 is not a member SAA7105H

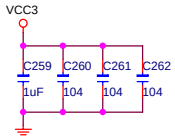
OPTION MOUNT

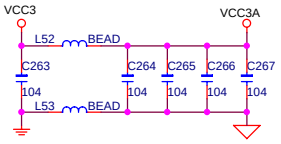
	LOCTION	LOCTION
OPTION1	U26	P16/ U28
OPTION2	P17/ U30	











	Enable TV-OUT	Disable TV-OUT
J4 Number	1,2	2,3

DMP ELECTRONICS INC.

Title	TV OUT SAA7105	
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LVDS Conn /SPI FLASH

8,14 FPD[23:0] — FPD[23:0]

8,14 FP1HS — FP1HS
8,14 FP1VS — FP1VS
8,14 FP1DE — FP1DE

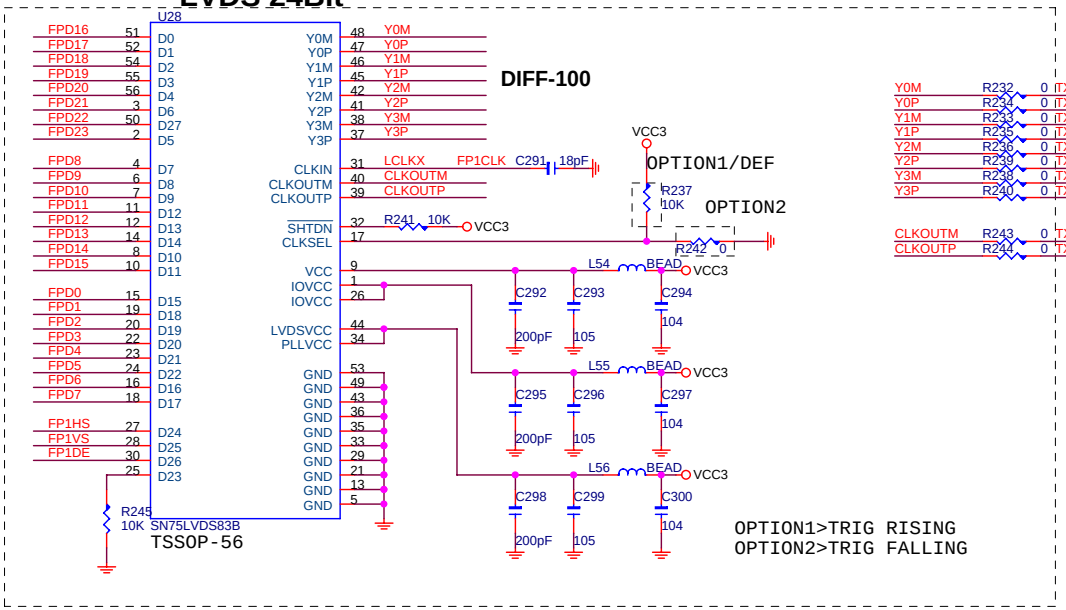
8,14 FP1CLK — FP1CLK

7 SPICS — SPICS
7 SPIDI — SPIDI
7 SPICLK — SPICLK
7 SPIDO — SPIDO

OPTION MOUNT

	LOCTION	LOCTION
OPTION1	U26	U28
OPTION2	P17/ U30	

LVDS 24Bit

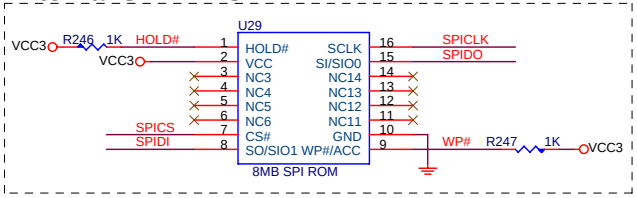


OPTION1/2

DVO definition table

Port Name/ DMP LCD pin	LVDS-24bit SDR	TTL-18bit	LVDS-12bit DDR+CAP	CAP+TV-12bit	DSTN	
					Color	Mono
FPD23	LVD23:R7	D17: R5				
FPD22	LVD22:R6	D16: R4	VSYNC	VSYNC		
FPD21	LVD21:R5	D15: R3	HSYNC	HSYNC		
FPD20	LVD20:R4	D14: R2	CAP_FILD	CAP_FILD		
FPD19	LVD19:R3	D13: R1			LD7	
FPD18	LVD18:R2	D12: R0			LD6	
FPD17	LVD17:R1				LD5	
FPD16	LVD16:R0				LD4	
FPD15	LVD15:G7	D11: G5			LD3	LD3
FPD14	LVD14:G6	D10: G4			LD2	LD2
FPD13	LVD13:G5	D09: G3			LD1	LD1
FPD12	LVD12:G4	D08: G2			LD0	LD0
FPD11	LVD11:G3	D07: G1	LV1D11:G3R7	TVD11		
FPD10	LVD10:G2	D06: G0	LV1D10:G2R6	TVD10		
FPD09	LVD09:G1		LV1D09:G1R5	TVD09		
FPD08	LVD08:G0		LV1D08:G0R4	TVD08		
FPD07	LVD07:B7	D05: B5	LV1D07:B7R3	TVD07	HD7	
FPD06	LVD06:B6	D04: B4	LV1D06:B6R2	TVD06	HD6	
FPD05	LVD05:B5	D03: B3	LV1D05:B5R1	TVD05	HD5	
FPD04	LVD04:B4	D02: B2	LV1D04:B4R0	TVD04	HD4	
FPD03	LVD03:B3	D01: B1	LV1D03:B3G7	TVD03	HD3	HD3
FPD02	LVD02:B2	D00: B0	LV1D02:B2G6	TVD02	HD2	HD2
FPD01	LVD01:B1		LV1D01:B1G5	TVD01	HD1	HD1
FPD00	LVD00:B0		LV1D00:B0G4	TVD00	HD0	HD0
FPD1DE/FPD2DE	LVDE	DE	LV1DE	TVDE	DISP	
FPD1HS/FPD2HS	LVHS	HS	LV1HS	TVHS	LOAD	
FPD1VS/FPD2VS	LVVS	VS	LV1VS	TVVS	FRM	
FPD1CLK/FPD2CLK	LVCLK	CLK	LV1CLK	TVCLK	CP	
TVCLKIN				TVCLKIN		
FPD1DET/FPD2DET	DVIDET	DVIDET	DVIDET	TVDET	DVIDET	
ENVEE	ENVEE	ENVEE	ENVEE		ENVEE	
ENPVDD	EPVDD	EPVDD	EPVDD		EPVDD	
ENBLT	ENBLT	ENBLT	ENBLT		ENBLT	
CAPD07			CAPD07	CAPD07		
CAPD06			CAPD06	CAPD06		
CAPD05			CAPD05	CAPD05		
CAPD04			CAPD04	CAPD04		
CAPD03			CAPD03	CAPD03		
CAPD02			CAPD02	CAPD02		
CAPD01			CAPD01	CAPD01		
CAPD00			CAPD00	CAPD00		
CAPCLK			CAPCLK	CAPCLK		

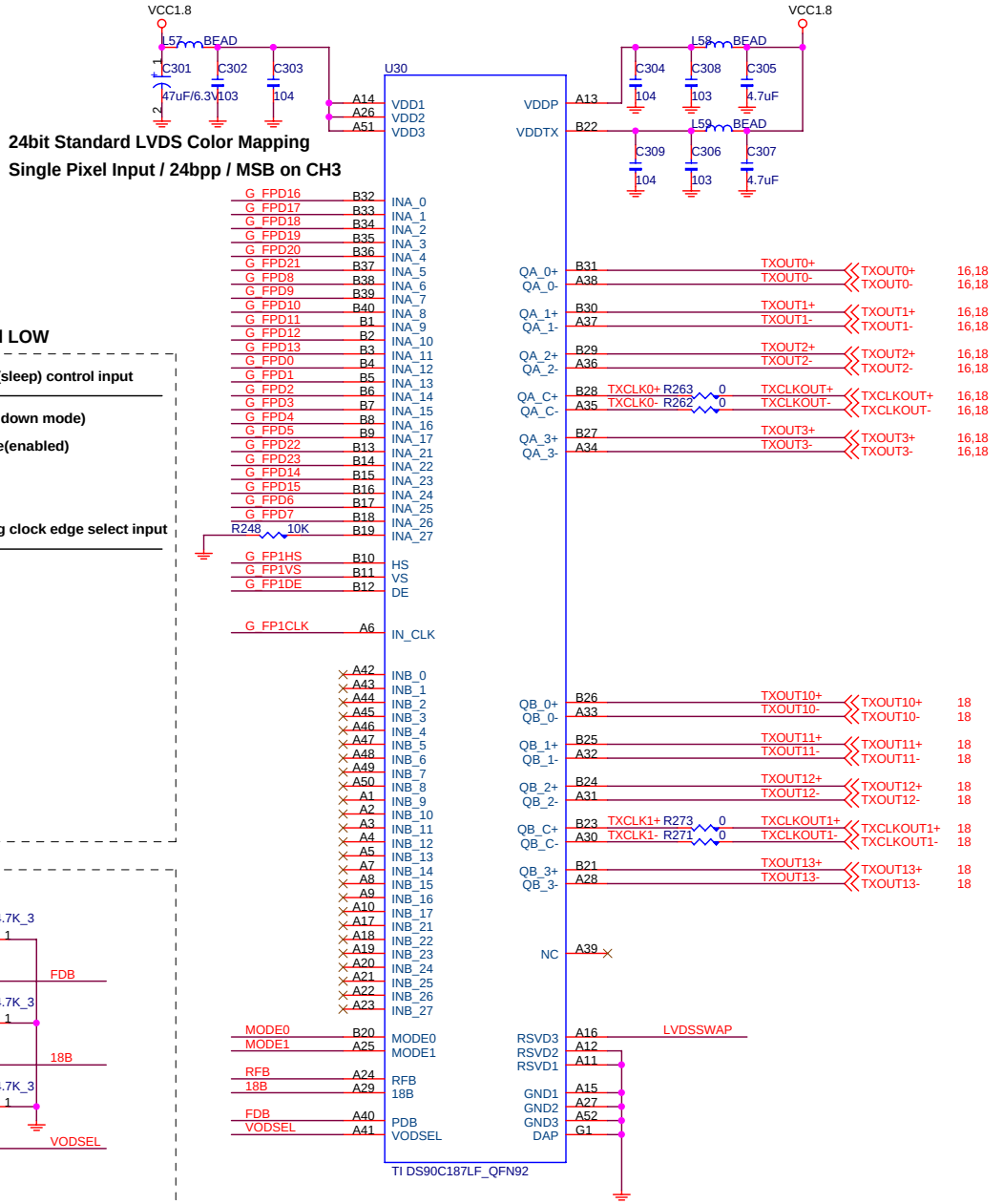
External SPI FLASH



DMP ELECTRONICS INC.

Title	LCD/LVDS		
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Dual Pixel FPD/LVDS_OPTION1/2



LVDSSWAP is reseved for future use and should be tied LOW

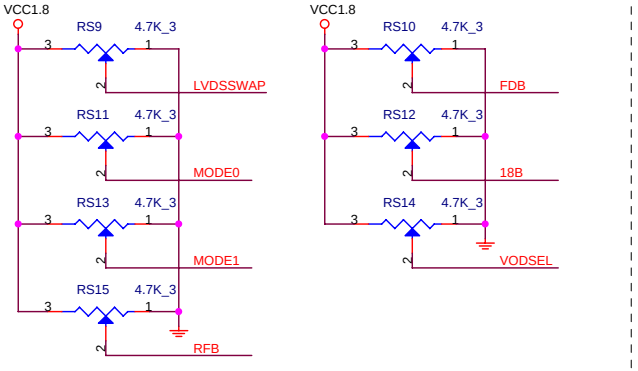
VODSEL	VOD Level select input
0(LOW)	Low swing
0(High)	Normal swing

FDB	Power down(sleep) control input
0(LOW)	Sleep(power down mode)
0(High)	Device active(enabled)

18B	18/24 bit control input
0(LOW)	Low swing
0(High)	Normal swing

RFB	Rising/Falling clock edge select input
0(LOW)	Falling edge
0(High)	Rising edge

Mode (1)	Mode (0)	Mode control inputs
0	0	Single in / Single out
0	1	Single in / Dual out
1	0	Dual in / Dual out
1	1	Reseved



DMP ELECTRONICS INC.	
Title	Dual Pixel FPD-LINK/LVDS
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Date:	Monday, February 02, 2015 — Sheet 17 of 19